ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/378331781

A Linear Programming Approach to Optimize the Storage Capacity

Conference Paper · December 2023

DOI: 10.1109/SMART59791.2023.10428501

CITATION
1

READS
164

3 authors:



Chhavi Gupta
Teerthanker Mahaveer University
4 PUBLICATIONS **5** CITATIONS
SEE PROFILE



Vipin Kumar
Teerthanker Mahaveer University
9 PUBLICATIONS **12** CITATIONS
SEE PROFILE



Kamesh Kumar
Teerthanker Mahaveer University
13 PUBLICATIONS **21** CITATIONS
SEE PROFILE

All content following this page was uploaded by Chhavi Gupta on 30 March 2024.

The user has requested enhancement of the downloaded file.

Proceedings of the SMART–2023, IEEE Conference ID: 59791
12th International Conference on System Modeling & Advancement in Research Trends, 22nd–23rd, December, 2023
College of Computing Sciences & Information Technology, Teerthanker Mahaveer University, Moradabad, India

A Linear Programming Approach to Optimize the Storage Capacity

Chhavi Gupta[1], Vipin Kumar[2] and Kamesh Kumar[3]
[1]*Research Scholar, Department of Mathematics, Faculty of Engineering, Teerthanker Mahaveer University, Moradabad, Uttar Pradesh, India*
[2]*Associate Professor, Department of Mathematics, Faculty of Engineering, Teerthanker Mahaveer University, Moradabad, Uttar Pradesh, India*
[3]*Assistant Professor, Department of Mathematics, Faculty of Engineering, Teerthanker Mahaveer University, Moradabad, Uttar Pradesh, India*
E-mail: [1] chhavi1official@gmail.com, [2]drvipink.engineering@tmu.ac.in, [3]drkamesh.engineering@tmu.ac.in

***Abstract*—In the present business landscape, enhancing warehouse efficiency can confer a competitive edge in the fiercely contested logistics market. This study proposes the application of mathematical programming to enhance the warehouse management process. The layout of the warehouse, incorporating various operational constraints, was identified for future enhancements. The initial phase of the research involves examining the current state of warehouse modeling practices, providing insights into the development of a linear programming model and identifying potential areas for improvement. The subsequent stage focuses on developing the linear programming model. Following this, a sensitivity analysis will be conducted to assess the impact of altering inventory policies.**

Keywords: Warehouse Management, Linear Programming (LP) Model, Inventory Policies.

I. Introduction

There are multiple rationales for maintaining inventory, with the paramount purpose being to mitigate the risks associated with fluctuations in demand and supply rates. [1] Inventory serves as a buffer between supply and demand, ensuring a company's ability to offer a high level of service crucial for its market survival. [2] Elevated inventory levels contribute to customer satisfaction and enable prompt responses to customer demands. However, [3] warehouse functions extend beyond inventory holding; frequently, warehouses facilitate seamless coordination throughout the entire supply chain by aiding in distribution, a particularly vital aspect for international corporations. [4, 5] Maintaining stocks and managing warehouses yield numerous advantages for a company, yet they also entail substantial costs. [6] Inventory costs rank among the most significant expenses within the realm of logistics. [7] The challenge lies in the fact that as inventory holding costs rise, achieving optimal warehouse management becomes increasingly crucial. [8] Effectively managing warehouses necessitates striking a balance between the benefits of stock retention and the associated expenses by attaining an appropriate inventory level. [9] Managers encounter the necessity to establish a suitable replenishment system that allows for the maintenance of a high service level while concurrently reducing inventory cost. [10] Considering only a few factors such as the large volume of daily customer orders, the influx of numerous supplies and products to the warehouse, the diverse nature of warehouse operations, the workforce size, and the unpredictable demand, devising a cost-effective replenishment policy appears to be intricate undertaking. [11] Employing mathematical modeling, particularly through linear programming, offers the potential to create supportive tools for enhancing replenishment policies and warehouse management. [12] Broadly, the aim of modeling is to enhance our understanding of reality and provide assistance in problem-solving. [13] Conclude in research paper that the LP technique furnishes leaders with the complete number of items that may be put away in a stockroom in view of the different variables considered in this review. Furthermore, [14] systems influencing the utilization of warehouse at full expense are obvious. [15] Production warehouses serve as storage for raw materials, semi-finished goods, and finished products within a production facility. [16] Various model types exist, but this paper specifically addresses mathematical models, focusing on linear programming modeling. [17] In addition to fostering improved comprehension, the primary objective of linear programming (LP) modeling is to attain an optimal solution. [18] The transition from a conceptual model to mathematical language involves expressing the model's goal as an objective function. [19, 20] Typically, this objective function seeks to maximize profits or minimize costs. Subsequently, [21] the constraints and specifications are translated into a system of equations and inequalities.

Copyright © IEEE–2023 ISBN: 979-8-3503-6988-5

II. AIM

The research's goal is to create a linear programming model aimed at enhancing the efficiency of warehouse management processes. To achieve this research objective, the following specific goals have been outlined:

- To discern the prevailing approaches in mathematical modeling for the management of warehouses.
- To ascertain the current state of warehouse management practices, with particular attention to replenishment policies.
- To create a linear programming model that aids the replenishment system.
- To validate the formulated linear programming model.
- To execute several managerial scenarios with the aim of enhancing the existing operational practices.

III. Methodology

The initial three objectives have been successfully achieved. To fulfill each objective, appropriate tools and techniques are outlined as follows.

In order to understand and identify the current warehouse management strategy, the semi-structured interviews with the warehouse manager & the process observation of warehouse operations supplied the necessary information. The subsequent phase involves gathering pertinent data from the warehouse. This includes a thorough examination of historical records and the capture of other quantitative data. These steps are essential for model development and result acquisition. The problem will be addressed using the Linear Programming model, represented through Excel Solver as the chosen methodology. It will also be employed in generating additional scenarios for result comparison. The development of more scenarios will utilize sensitivity analysis as the chosen methodology.

IV. Numerical Approach

The warehouse receives regular shipments from external suppliers and engages in the acquisition and resale of products, ensuring a suitable profit margin. Its customer base primarily consists of networks of small product consumers. Additionally, the warehouse extends its services to third-party logistics, holding inventory on behalf of external companies. The warehouse operates around the clock, dividing the workload into three eight-hour shifts for continuous 24-hour coverage. The warehouse encompasses four key processes: Goods-in involves receiving, storage takes place in the designated storage area, and the Receiving division encompasses unloading products, verifying product counts for accuracy, updating the warehouse management system with the latest inventory levels, and allocating goods to suitable storage locations & also evaluate that how many pallets are necessary to store in other warehouse. For this purpose we use linear programming model (LPM) to evaluate the problem of space optimization. All pertinent parameters, a decision variable, an objective function, and constraints are included in the formulation. The resulting solution will be presented as follows.

Table 1: Categorize the product groups based on the dimensions of pallets

Group of pallets	Length (m)	Width(m)	Height (m)	Volume (m^3)	Amount of Pallet
1	1.18	1.2	1.4	1.9824	14
2	1.3	0.9	1.2	1.404	11
3	1.2	1.2	0.7	1.008	13
4	1.2	1.2	1.4	2.016	12
5	0.83	1.05	0.86	0.749	15

It illustrates the capacity of each pallet group that can be accommodated in a single truck. Nevertheless, it is also possible to store a considerable quantity of pallets from various groups in a single truck, with an average of approximately 14 mixed pallets per truck round. This is primarily due to the fact that the automotive spare parts, categorized under WH 1 and WH 2 (checking part) in Figure 2, will be allocated to specific groups, allowing them to be stored collectively.

Main parameter:

m = total quantity of pallet's group

Pi = The size of pallet dimensions within group i (where i = 1,……,n)

Qi = The quantity of pallets accessible in group i (where i = 1,……,n)

Di = The lowest allowable pallet capacity within group i (where i = 1,…..,n)

N = Storage space designated for automotive spare parts.

V. Model Formulation

Decision variable:

Yi = The quantity of pallets moved within group i.(where i = 1,……,n)

Objective Function:

$$Maximize\ Z = \sum_{i=1}^{n} P_i Y_i$$

Constraints:

1. The storage capacity for automotive spare parts in the new warehouse must accommodate an equal or smaller space for relocated pallets.

$$\sum_{i=1}^{n} P_i Y_i \leq N \qquad (where\ i = 1, \ldots n)$$

Copyright © IEEE–2023 ISBN: 979-8-3503-6988-5

2. The quantity of relocated pallets should match or exceed the minimum capacity that can be stored in a single round of truck.
   $Y_i \leq D_i \quad (where\ i = 1, \dots n)$
3. The quantity of relocated pallets should be the equal or less than the existing stock of pallets prior to relocation.
   $Y_i \leq Q_i \quad (where\ i = 1, \dots n)$
4. The quantity of relocated pallets is a positive integer.
   $Y_i \leq 0 \quad (where\ i = 1, \dots n)$

VI. Illustrating a Model for Optimization

Objective function:

$$Maximize\ Z = 1.9824Y_1 + 1.404Y_2 + 1.008Y_3 + 2.016Y_4 + 0.749Y_5 \quad [1]$$

Constraints:

$$1.9824Y_1 + 1.404Y_2 + 1.008Y_3 + 2.016Y_4 + 0.749Y_5 \leq 1700 \quad [2]$$

$$Y_1 \geq 14,\ Y_2 \geq 11,\ Y_3 \geq 13,\ Y_4 \geq 12,\ Y_5 \geq 15 \quad [3]$$

$$Y_1 \leq 780, Y_2 \leq 95, Y_3 \leq 86, Y_4 \leq 120, Y_5 \leq 196 \quad [4]$$

Non negativity constraints:

$$Y_1, Y_2, Y_3, Y_4, Y_5 \geq 0$$

After evaluating the total volumes for each pallet group & completing the setup of a linear programming (LP), we execute a simulation using the Excel solver. The result will be presented that the new warehouse is equipped to handle a maximum storage volume of 1,700 m^3 for automotive spare-parts pallets. Additionally, the minimum pallet capacity within each group will match the proportion of categorized pallets that can be accommodated in a single truck round. The supply of available pallets will be determined by the maximum demand of each group before the relocation to the new warehouse.

VII. Conclusion

Once the optimization model is completed, we contrasts the outcomes between the current solution & a novel solution utilizing the linear programming method. We analyze the previous study of warehouse management in which total volume of a warehouse was 1400m^3 and the storage area of new warehouse is 1700m^3. This finding reveal that the new warehouse has more storage capacity. The availability of pallets is contingent upon the maximum demand of each group the relocation to the new warehouse. The findings indicate that if the company continues to determine the number of pallets using the existing solution. The company utilizes the linear programming (LP) method to compute the maximum capacity for relocated pallets, the issue of excessive pallet position will diminish as the compatibility of the relocated pallet quantity with the storage area in the new warehouse improves.

References

[1] Manzini, R., Accorsi, R., Pattitoni, L., & Regattieri, A. (2011). A Supporting Decisions Platform for the Design and Optimization of a Storage Industrial System. Efficient Decision Support Systems - Practice and Challenges in Multidisciplinary Domains, September. https://doi.org/10.5772/16861

[2] Saxena, A. K. (2021, December). Review on EOQ Models for Instanteneous and Non-Instanteneous Deteriorating Items. In *2021 10th International Conference on System Modeling & Advancement in Research Trends (SMART)* (pp. 312-315). IEEE. 10.1109/SMART52563.2021.9676264

[3] Kumar, P., Saxena, A., Gupta, G. K., & Kumar, K. (2022, December). Duality of Graph and Its Application in Electric Circuit. In 2022 11th International Conference on System Modeling & Advancement in Research Trends (SMART) (pp. 1561-1563). IEEE.

[4] Gupta, C., Kumar, V., & Kumar, K. (2022, December). A Study on the Applications of Supply Chain Management. In *2022 11th International Conference on System Modeling & Advancement in Research Trends (SMART)* (pp. 737-741). IEEE. 10.1109/SMART55829.2022.10047167

[5] Saxena, A. K. (2022, December). An EOQ Model for Deteriorating Items with Time Dependent Demand. In 2022 11th International Conference on System Modeling & Advancement in Research Trends (SMART) (pp. 926-930). IEEE. 10.1109/SMART55829.2022.10046704

[6] Perera, D., Fernando, A., & Mirando, U. (2021). Warehouse Space Optimization Using a Linear Programming Model.

[7] Perera, D., Mirando, U., & Fernando, A. (2022). WAREHOUSE SPACE OPTIMIZATION USING LINEAR PROGRAMMING MODEL AND GOAL PROGRAMMING MODEL.

[8] Van den Berg, J. P., & Zijm, W. H. (1999). Models for warehouse management: Classification and examples. *International journal of production economics*, *59*(1-3), 519-528.

[9] Dantzig, G. (1963). *Linear programming and extensions*. Princeton university press.

[10] Kantasa-ard, A. (2017). Optimization of the number of relocated automotive spare-part pallets using linear programming model. In *Proceedings of the 3rd RMUTT Global Business and Economics International Conference* (pp. 25-26).

[11] Kumar, V., & Saxena, A. (2012). Evaluation of High Speed Network like a Synchronous Transfer Mode. *International Journal of Applied Physics and Mathematics*, *2*(4), 253.

[12] Panda, S., Saha, S., & Basu, M. (2009). An EOQ model for perishable products with discounted selling price and stock dependent demand. *Central European Journal of Operations Research*, *17*, 31-53.

[13] Gupta, C., Kumar, V., & Gola, K. K. (2022, December). Implementation Analysis for the Applications of Warehouse Model Under Linear Integer Problems. In *International Conference on Intelligent Systems Design and Applications* (pp. 239-247). Cham: Springer Nature Switzerland. 10.1007/978-3-031-35507-3_24

[14] Pohl, L. M., Meller, R. D., & Gue, K. R. (2009). An analysis of dual-command operations in common warehouse designs. Transportation Research Part E: Logistics and Transportation Review, 45(3), 367–379. https://doi.org/10.1016/j.tre.2008.09.010

[15] Karasek, J. (2013). An Overview of Warehouse Optimization. International Journal of Advances in Telecommunications, Electrotechnics, Signals and Systems, 2(3). https://doi.org/10.11601/ijates.v2i3.61

 Copyright © IEEE–2023 ISBN: 979-8-3503-6988-5

[16] Geraldes, C. A. S., Carvalho, M. S., & Pereira, G. A. B. (2012). Warehouse design and product assignment and allocation: A mathematical programming model. ESM 2012 - 2012 European Simulation and Modelling Conference: Modelling and Simulation 2012, 314–320.

[17] Muppani (Muppant), V. R., & Adil, G. K. (2008). Efficient formation of storage classes for warehouse storage location assignment: A simulated annealing approach. Omega, 36(4), 609–618. https://doi.org/10.1016/j.omega.2007.01.006

[18] Singh, V. D., Ali, S. R., & Piyoosh, A. K. (2022, December). A Review on the Relationship between LULC and LST using Geospatial Technologies. In 2022 11th International Conference on System Modeling & Advancement in Research Trends (SMART) (pp. 472-476). IEEE.

[19] Dubey, R., Singh, T., & Tiwari, S. (2012). Supply Chain Innovation is a Key to Superior Firm Performance an Insight from Indian Cement Manufacturing. International Journal of Innovation Science, 4(4), pp. 217-230.

[20] Al Amin, M., Das, A., Roy, S., & Shikdar, M. I. (2019). Warehouse selection problem solution by using proper mcdm process. *Int. J. Sci. Qual. Anal*, *5*(2), 43-51.

[21] Zuñiga, C., Angel Piera, M., & Narciso, M. (2011). Revisiting the pallet loading problem using a discrete event system approach to minimise logistic costs. *International Journal of Production Research*, *49*(8), 2243-2264.

[22] Khyati, Ashendra Kumar Saxena, "An EOQ Model for Deteriorating Items with Time Dependent Demand", 2022 11th International Conference on System Modeling & Advancement in Research Trends (SMART), pp.926-930, 2022.

[23] Saxena, A. K., & Yadav, R. K. (2011, December). Ordering Policy For Non Instantaneous Decaying Items With Stock Dependent Demand and Shortages. In Proceedings of the International Conference on Soft Computing for Problem Solving (SocProS 2011) (pp. 497-513).

Copyright © IEEE–2023 ISBN: 979-8-3503-6988-5

View publication stats